ETSI TS 103 636-1 V1.2.1 (2021-04)



DECT-2020 New Radio (NR);
Part 1: Overview;
Release 1

Reference
RTS/DECT-00361

Keywords
5G, DECT, DECT-2020, IMT-2020, NR, OFDM, radio

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from:
http://www.etsi.org/standards-search

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format at www.etsi.org/deliver.

Users of the present document should be aware that the document may be subject to revision or change of status.
Information on the current status of this and other ETSI documents is available at
https://portal.etsi.org/TB/ETSIDeliverableStatus.aspx

If you find errors in the present document, please send your comment to one of the following services:
https://portal.etsi.org/People/CommiteeSupportStaff.aspx

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2021.
All rights reserved.

Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
1 Scope 5
2 References 5
2.1 Normative references 5
2.2 Informative references 5
3 Definition of terms, symbols and abbreviations 5
3.1 Terms 5
3.2 Symbols 6
3.3 Abbreviations 6
4 General 7
4.1 Introduction 7
4.2 Overview of the parts of DECT-2020 Technical Specifications 8
5 System and Network Architectures 8
5.1 Wireless Point-to-Point and Point-to-Multipoint Links 8
5.2 Local Area Wireless Access Networks in Cellular Network Topology 9
5.3 Mesh network topology 9
5.3.1 Introduction 9
5.3.2 Mesh system operation 9
5.3.3 Mesh Routing 11
5.4 Interworking 12
6 Overview on Radio technology 13
6.1 Radio interface protocol architecture 13
6.2 Physical Layer 13
6.2.1 Physical Layer functions and capabilities 13
6.2.2 Radio characteristics 14
6.3 Medium Access Control Layer 15
6.3.1 Introduction 15
6.3.2 MAC Structure 15
6.3.3 Identities 16
6.3.4 Services 17
6.3.5 Functions 17
6.3.6 Channel Structure 18
6.4 Mobility and State Transitions 18
6.4.1 Overview 18
6.4.2 Intra DECT-2020 Mobility 19
History 20

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (https://ipr.etsi.org/).

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™** and **LTE™** are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Digital Enhanced Cordless Telecommunications (DECT).

The present document is part 1 of a multi-part deliverable covering the overall description of DECT-2020 NR standards, as identified below:

Part 1: "Overview";

Part 2: "Radio reception and transmission requirements";

Part 3: "Physical layer";

Part 4: "MAC layer".

The present document introduces the system overview covering massive Machine Type Communication (mMTC) and Ultra-Reliable Low Latency Communication (URLLC) features.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document provides an overview on DECT-2020 NR including layers, system and network architectures envisioned for this release. Further it provides an overview to ETSI TS 103 636-2 [1], ETSI TS 103 636-3 [2], ETSI TS 103 636-4 [3] and their interrelation.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 103 636-2: "DECT-2020 New Radio (NR); Part 2: Radio reception and transmission requirements; Release 1".

[2] ETSI TS 103 636-3: "DECT-2020 New Radio (NR); Part 3: Physical layer; Release 1".

[3] ETSI TS 103 636-4: "DECT-2020 New Radio (NR); Part 4: MAC layer; Release 1".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI TS 123 501: "5G; System architecture for the 5G System (5GS) (3GPP TS 23.501 Release 16)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

Fixed Termination point (FT): operational mode of RD where RD initiates coordinates local radio resources, provides information how other RDs may connect and communicate with it

operating channel: single continuous part of radio spectrum with a defined bandwidth where RDs transmits and/or receives

Portable Termination point (PT): operational mode of RD where RD selects another RD, which is in FT mode, for association

Radio Device (RD): device with radio transmission and reception capability, which can operate in FT and/or PT mode

resource: variable length time unit defined in subslot(s) or slot(s) in single operating channel that RD is using for transmission or reception of physical layer packet

NOTE: Resource can be contentious or contention free, i.e. scheduled.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

RD_{FT}	RD operating in FT mode
$RD_{FT,PT}$	RD operating in both FT and PT mode
RD_{PT}	RD operating in PT mode

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ARQ	Automatic Repeat reQuest
BCC	Broadcast Control
BCCH	Broadcast Control Channel
BLER	Block Error Ratio
BPSK	Binary Phase Shift Keying
BSC	Beacon Scanning Control
CCC	Connection Configuration Control
CCCH	Common Control CHannel
CP-OFDM	Cyclic Prefix Orthogonal Frequency Division Multiplexing
CRC	Cyclic Redundancy Check
DCCH	Dedicated Control CHannel
DCH	Dedicated CHannel
DECT	Digital Enhanced Cordless Telecommunications
DL	Downlink
DTCH	Dedicated Traffic CHannel
FDMA	Frequency Division Multiple Access
FEC	Forward Error Correction
FFT	Fast Fourier Transform
FP	Fixed Part
FT	Fixed Termination point
GI	Guard Interval
HARQ	Hybrid Automatic Repeat Request
ID	IDentity
IMT	International Mobile Telecommunications
IoT	Internet of Things
ITU-R	International Telecommunication Union-Radiocommunication sector
LRC	Local Radio Control
LSB	Least Significant Bit
MAC	Medium Access Control
MCS	Modulation and Coding Scheme
MIMO	Multiple Input Multiple Output
mMTC	massive Machine Type Communication
MSB	Most Significant Bit
MTCH	Multicast (Broadcast) Traffic Channel
N3IWF	Non-3GPP Inter-Working-Function
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
PCC	Physical Control Channel
PCCH	Paging Common Channel

PCH/BCH	Paging and Broadcast channel
PDC	Physical Data Channel
PDU	Protocol Data Unit
PHY	Physical Layer
PLMN	Public Land Mobile Network
PT	Portable Termination point
PTC	Paging Transmission Control
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RAC	Random Access Control
RACH	Random Access CHannel
RD	Radio Device
RF	Radio Frequency
RIT	Radio Interface Technology
RSSI	Received Signal Strength Indicator
RX	Receiver
RX-TX	Receive-Transmit
TDD	Time Division Duplex
TDMA	Time Division Multiple Access
TNGF	Trusted Non-3GPP Gateway Function
TX	Transmitter
UE	User Equipment
UL	Uplink
ULE	Ultra Low Energy
URLLC	Ultra-Reliable Low Latency Communication
WAN	Wide Area Networks

4 General

4.1 Introduction

DECT-2020 NR is a Radio Interface Technology (RIT) designed to provide a slim but powerful technology foundation for wireless applications deployed in various use cases and markets.

This radio technology includes, but is not limited to Cordless Telephony, Audio Streaming Applications, Professional Audio Applications, consumer and industrial applications of Internet of Things (IoT) such as industry and building automation and monitoring, and in general solutions for local area deployments for Ultra-Reliable Low Latency Communication (URLLC) and massive Machine Type Communication (mMTC) as envisioned by ITU-R for IMT-2020.

In general, DECT-2020 NR as a technology foundation is targeted for local area wireless applications, which can be deployed anywhere by anyone at any time. The technology supports autonomous and automatic operation with minimal maintenance effort. Where applicable, interworking functions to Wide Area Networks (WAN). e.g. PLMN, satellite, fibre, and internet protocols foster the vision of a network of networks.

DECT-2020 NR can be used as a foundation for:

- Very reliable Point-to-Point and Point-to-Multipoint Wireless Links provisioning (e.g. cable replacement solutions);
- Local Area Wireless Access Networks following a star topology as in classical DECT deployment supporting URLLC use cases; and
- Self-Organizing Local Area Wireless Access Networks following a mesh network topology, which enables to support mMTC use cases.

DECT-2020 NR applies similar design principles as in legacy DECT and DECT ULE. Especially the inherent feature of automatic interference management allows deployments without extensive frequency planning. The Mesh networking capability of DECT-2020 NR enables application-driven network topologies and deployments in e.g. IoT and mMTC use scenarios such that the link budget of classical cellular base-station to user equipment constellations is no longer a limiting factor.

The DECT-2020 NR physical layer is in principle is suited to addressing frequency bands below 6 GHz. The physical layer employs Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) combined with Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) in a Time Division Duplex (TDD) communication manner. The physical layer employs multiple numerologies, with different subcarrier spacings and corresponding Cyclic Prefix lengths and FFT sizes, allowing operation with different channel bandwidths, and optimize operations in different frequency bands and propagation environments. The physical layer supports advanced channel coding (Turbo coding) for both control and physical channels and Hybrid ARQ with incremental redundancy, which enables fast re-transmission. Advanced channel coding together with Hybrid ARQ ensures very reliable communication.

Additionally, the physical layer supports, fast link adaptation, transmit and receiver diversity, as well as MIMO operations up to 8 streams.

DECT-2020 NR (i.e. PHY layer numerology and MAC algorithms) is designed to enable coexistence with legacy DECT and DECT evolution in current frequency bands allocated to DECT.

4.2 Overview of the parts of DECT-2020 Technical Specifications

Release 1 of the DECT-2020 NR technical specifications defines the Radio Interface Technology (RIT) by the following parts:

- ETSI TS 103 636-1 (the present document): "DECT-2020 New Radio (NR); Part 1: Overview".
- ETSI TS 103 636-2: "DECT-2020 New Radio (NR); Part 2: Radio Reception and Transmission requirements" [1].
- ETSI TS 103 636-3: "DECT-2020 New Radio (NR); Part 3: Physical layer" [2].
- ETSI TS 103 636-4: "DECT-2020 New Radio (NR); Part 4: Medium Access Control layer" [3].

ETSI TS 103 636 series will be accompanied by a feature and/or application-driven technical specification set, which is organized as a multi-part deliverable, delivering profiles and application specific solutions for various industries.

ETSI TS 103 636-1 is the present document.

ETSI TS 103 636-2 [1] establishes the minimum RF requirements for DECT-2020 New Radio (NR) Radio Devices (RDs). These requirements cover both Fixed Termination point (FT) as well as Portable Termination point (PT). This document also provides a list of supported frequency bands.

ETSI TS 103 636-3 [2] specifies the physical layer (PHY) and interaction between PHY and MAC layer.

ETSI TS 103 636-4 [3] specifies MAC layer and interaction between MAC layer and physical layer and higher layers.

5 System and Network Architectures

5.1 Wireless Point-to-Point and Point-to-Multipoint Links

Wireless Point-to-Point links involve two radio devices communicating with each other. A typical application is the cable replacement by a wireless link established between two radio devices requiring communicating with each other.

Compared to wireline systems, wireless comes with the benefit that point to multipoint communication is an inherent feature of radio propagation, so that the support of broadcast and multicast messages from one point to multiple points is just a matter of protocol.

The radio connection between two or more radio devices is enabled by one RD selecting to operate in FT mode (RD_{FT}) and initiate radio resource coordination and beacon transmissions. Other RD(s) perform association procedure in PT mode (RD_{PT}) with the RD_{FT}.

5.2 Local Area Wireless Access Networks in Cellular Network Topology

A single-cell network topology involves in principle two types of Radio Devices (RDs): an RD operates in FT mode (RD_{FT}) as a base station, which is a component of the fixed network infrastructure, other RDs operate PT mode (RD_{PT}).

RD_{FT} is coordinating radio resources, and serves a communication cell by being the central communication point for, RD_{PT}, which can be portable device.

A multi-cell topology is a deployment of multiple RD_{FT} as base stations in a fixed network infrastructure, where each base station is serving its own dedicated cell area and RD_{PT} can move from one cell area to the other.

5.3 Mesh network topology

5.3.1 Introduction

In DECT-2020 mesh network devices can communicate directly to each other extending the range of network and increasing the reliability of communication. The mode of the involved radio devices may change autonomously depending the context of the communication. Each radio device can act as a node transmitting a message, as a node forwarding any message from another radio device or as a node being the destination of a message. Each radio device can communicate directly (device to device) or, if not in range, indirectly - via other radio devices establishing a communication route - with each other which minimizes the probability of outage.

Mesh topology can support high device densities and the autonomous routing provides the ability to adapt dynamically mobile users and interference.

Mesh operation supports autonomous routing. In order to achieve efficient mMTC operation the mesh system is scalable to a very high number of devices in a network, the routing is based on cost value, without the need to maintain routing tables in each device.

The key requirements of how the scalability can be achieved are:

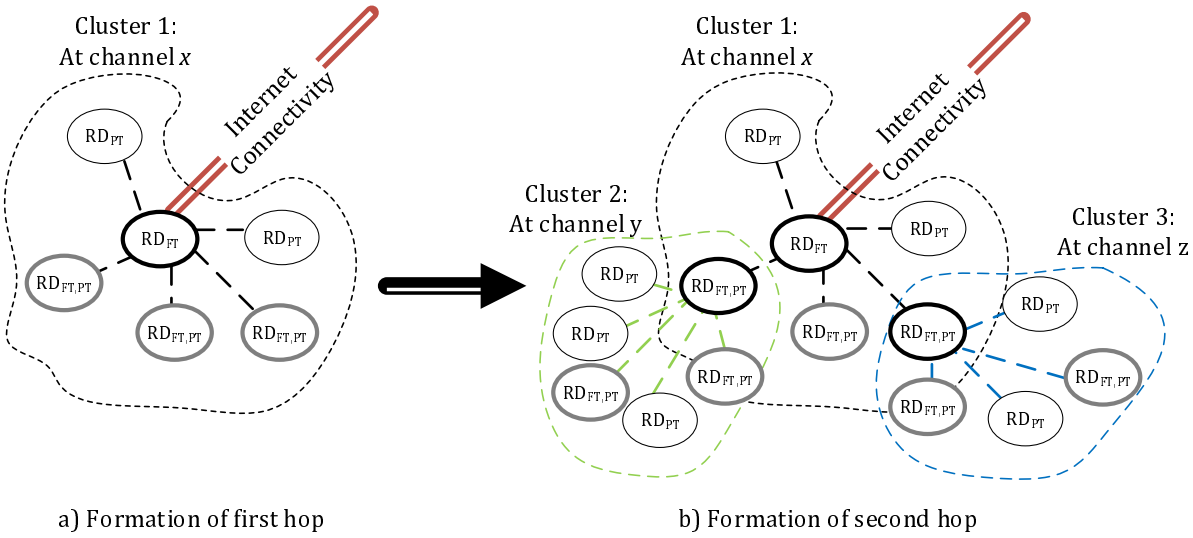
- All radio devices can route data. Whether RD is routing data is based on an autonomous decision of the RD. In addition, an RD may be configured to operate in PT mode only, e.g. due to low battery resources.
- Radio devices take local decisions of the radio recourses, e.g. how radio devices use Hybrid ARQ, selects modulation and coding and so forth in each radio link.
- Radio devices may change their operating mode between FT mode (RD_{FT}), PT mode (RD_{PT}), or both FT and PT modes ($RD_{FT,PT}$), autonomously based on local decisions.
- No central coordinator(s), enabling the massive scale of the network.
- Radio device operating in RD_{FT} or $RD_{FT,\ PT}$ mode coordinates local radio resouces.
- Support of multiple backend connected Radio devices that operate in FT mode (RD_{FT}).
- RDs can operate with multiple radio channels.

5.3.2 Mesh system operation

The mesh system operation is based on a clustered tree topology where each RD decides the next hop individually based available routes towards the RD providing the connection to the external internet in FT mode (RD_{FT}). Each radio device has knowledge of the next uplink and downlink hop in the clustered tree and RD_{FT}, or $RD_{FT,PT}$ mode in each cluster controls radio resources and transmissions independently.

The formation of clustered tree topology has following steps:

- An RD which has internet connectivity, RD_{FT}, in FT mode i.e. *Sink* selects operating frequency (or frequencies) and initiates a beacon transmission indicating that it has a route to the external world. This enables other RDs to detect it and associate with it. Beacons indicate all necessary parameters how to perform association, such as frame timing and how radio resources are used and the set of routing parameters. This association procedure does not differentiate from the association process in other system architectures described in clauses 5.1 and 5.2.
- RD detecting a beacon from another RD evaluates the connection based on the information included in the received beacon. Based on the information and signal quality the RD does an independent decision to which RD_{FT} or $RD_{FT,PT}$ to associate. RD monitors its neighbourhood and may autonomously initiate an association process towards another RD based on routing cost.
- Process continues to next hops and so on and it is illustrated in Figure 5.3.2-1.



NOTE 1: At the formation of the first hop a) one or more RD_{PT} or $RD_{FT,PT}$ associates with an RD_{FT}.
NOTE 2: The second hop b) is formed by RD_{PT} or $RD_{FT,PT}$ that associate with the first hop $RD_{FT.PT}$.
NOTE 3: Black thick circle: RD with associated members, grey thick circle: RD that is available routing, but yet has no associated members, black thin circle: RD in PT only mode.

Figure 5.3.2-1: Formation of the clustered tree mesh network topology

Once the RD has connectivity to the next hop, it can start sending data towards the RD in FT-mode. It can simply use a specific address value to indicate that the data is addressed to a backend. A backend system or other RDs can send DL data to the associated RD.

The basic beaconing and association procedures between RDs in mesh topology and in star topology are same. To enable battery powered RDs which are capable for routing data, the beacon transmission interval can be set longer.

Beaconing intervals extend to several seconds depending on the actual use case enabling low power RDs and routing RDs operations. On the other hand, when RD is mains powered, the beaconing period could be more frequent.

The system operation with multiple RDs in FT mode (*sinks*) is illustrates in Figure 5.3.2-2. The process of forming clusters is identical and a RD may choose to change its association to the next hop RD regardless if the next hop RD will provide connectivity to a different RD_{FT} having the backend connection. Figure 5.3.2-2 also illustrates the case when there is $RD_{FT,PT}$ that does not yet have any associated RD_{PT}.

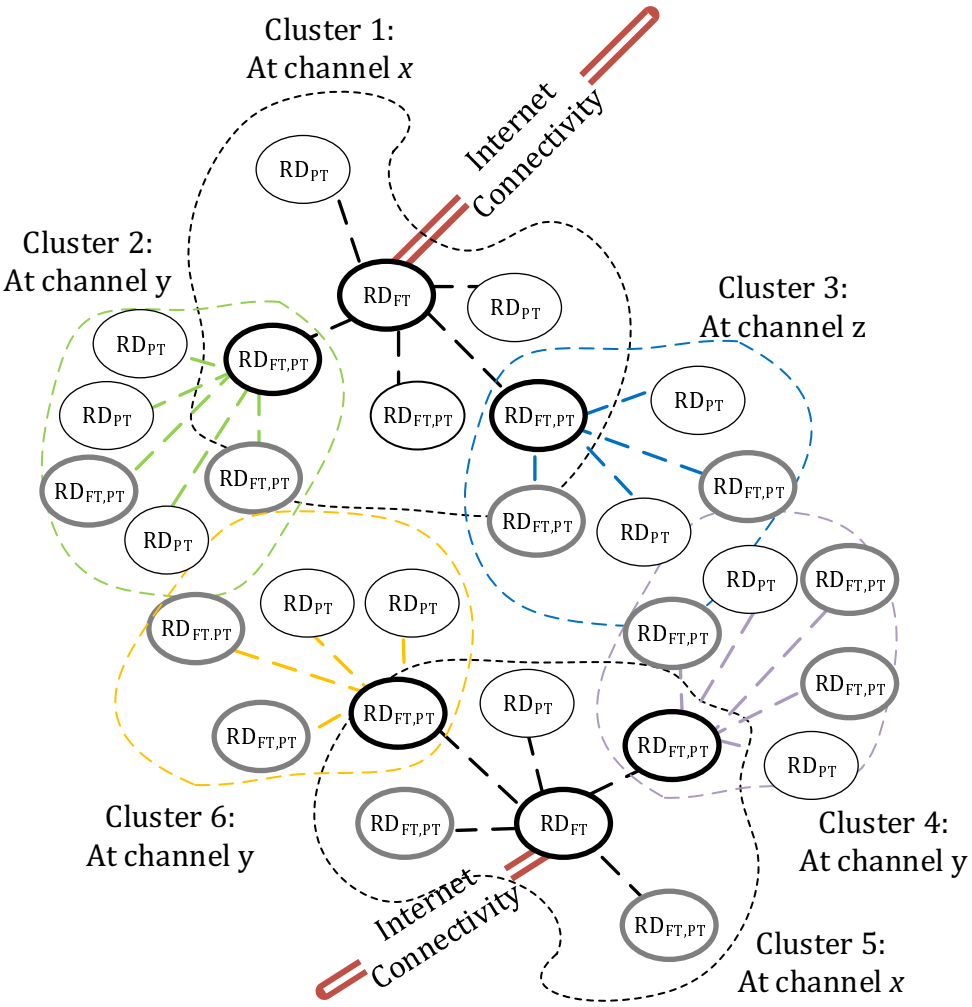


Figure 5.3.2-2: Example formation of the clusters in multiple RD_{FT} scenario

5.3.3 Mesh Routing

RD associates to the next hop RD based on received beacons quality attributes such as Received Signal Strength Indicator (RSSI) measurement from a beacon signal which assist it to determine the pathloss. If both DL pathloss and UL pathloss are such that connection can be established, the RD transmitting this beacon is considered as *potential next hop*. If RD detects multiple beacons from multiple RDs that meet the criteria of reliable connection, the RD considers the *route cost.* The *route cost* expresses the cost of the route to deliver data to the *sink*, i.e. to the RD_{FT} that has a backend connection to the internet. RD may select the RD_{FT} or $RD_{FT,PT}$ for association which indicates the smallest *cost*.

How the RD calculates the value for the *route cost* is left to implementation, as detailed the calculation can be dependent on multiple factors, such as RD capabilities, data rate, interference and BLER, own load i.e. data amount to be delivered, available battery energy, etc. Route cost value will increase at least 1 in every hop. The maximum route cost value is 254, which is not seen as a limiting factor in real deployments.

Routing has also a *sink address*, which is the long RD ID (32 bits) of the RD_{FT} having the backend connection. When a system supports multiple RD_{FT}s, (the actual number of RD_{FT}s can be significantly large), the RD may send short packet to the backend so that that backend can update its knowledge of where different RDs are located in the mesh network and send DL traffic only to the correct RD_{FT}.

The third routing parameter, *application sequence number*, provides identification for network level application data that needs to be distributed in the DL direction to all members of the network. The application sequence number is used by the RD associated to the next hop to identify whether the application data has changed compared to the current application data. If the sequence number is increased the RD requests the application data from its next hop.

The actual application data is outside of the scope of DECT-2020 NR but it can contain e.g. configuration data of the RD application such as measurement quantities and corresponding reporting frequencies, etc.

5.4 Interworking

The interworking with 3GPP 5G networks is supported by DECT-2020 NR via interworking profiles.

DECT-2020 can be connected as untrusted non-3GPP access network to the 5G Core Network via a Non-3GPP Inter-Working-Function (N3IWF) or as a trusted non-3GPP access network via a Trusted Non-3GPP Gateway Function (TNGF).

Figure 5.4-1 and Figure 5.4-2 provide an overview on the network architecture restricted to the relevant entities. For details refer to ETSI TS 123 501 [i.1].

A DECT-2020 FT can for example act as an intelligent gateway to the 3GPP network. In the case of untrusted Non-3GPP access the DECT-2020 FP is connected via N3IWF, while in case of trusted Non-3GPP access the DECT-2020 FT connects via TNGF.

A DECT-2020 PT if it supports both technologies can act as well as UE of the 3GPP network. Depending on capabilities of the DECT-2020 PT can be part of both networks in parallel or on demand.

DECT-2020 provides "standalone" non-3GPP accesses to the 5G Core Network, so a DECT-2020 network is operated fully independent and outside of the 3GPP NG-RAN.

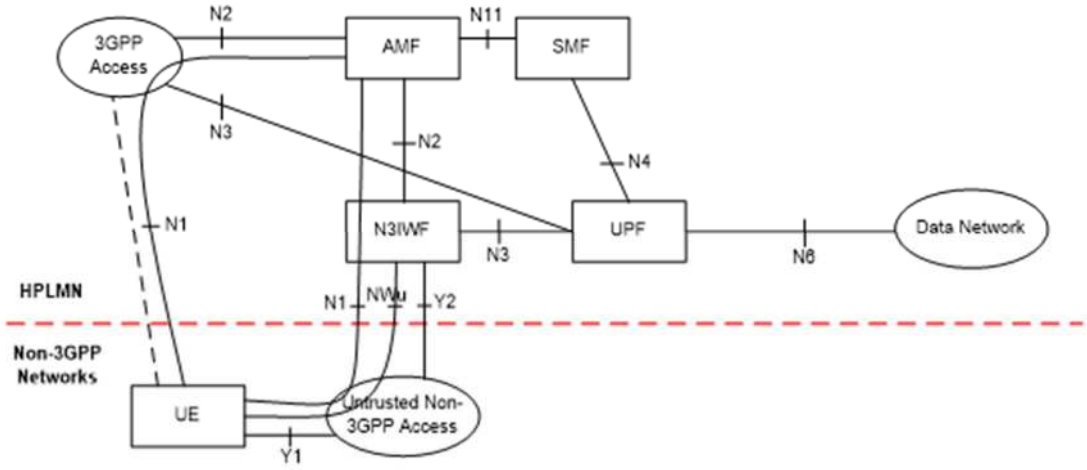


Figure 5.4-1: Non-roaming architecture for 5G Core Network with untrusted non-3GPP access [i.1]

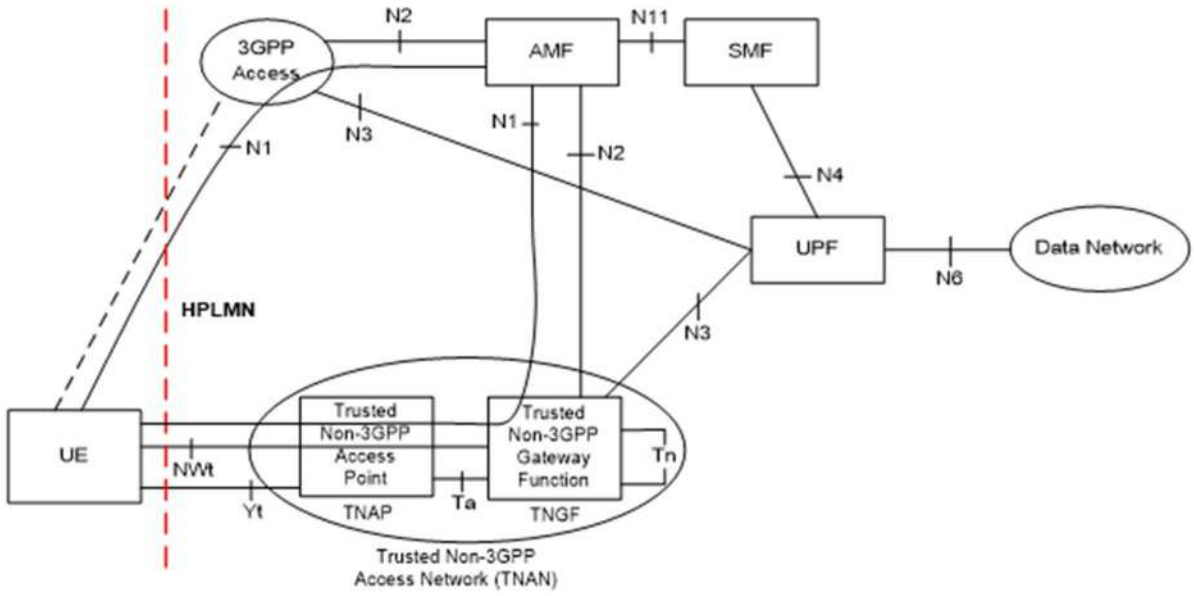


Figure 5.4-2: Non-roaming architecture for 5G Core Network with trusted non-3GPP access [i.1]

6 Overview on Radio technology

6.1 Radio interface protocol architecture

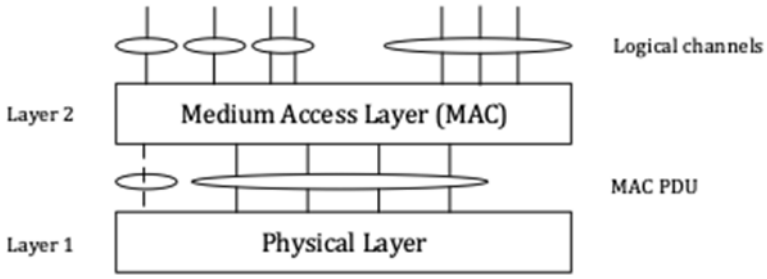


Figure 6.1-1: Radio interface protocol architecture

Figure 6.1-1 shows the DECT-2020 radio interface protocol architecture around the physical layer (PHY). The physical layer interfaces the Medium Access Control (MAC) layer. The circles between different layer/sub-layers indicate Service Access Points (SAPs). The physical layer offers Physical Control Channel (PCC) and Physical Data Channel (PDC) to transmit MAC PDU(s). Different physical channels are characterized by how the information is transferred over the radio interface within a single transmission packet.

6.2 Physical Layer

6.2.1 Physical Layer functions and capabilities

The DECT-2020 NR physical layer is in principle suitable to address frequency bands below 6 GHz. The physical layer employs Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) combined with Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) in a Time Division Duplex (TDD) communication manner.

Subcarrier spacing is defined by the subcarrier scaling factor μ, resulting either in 27 kHz, 54 kHz, 108 kHz or 216 kHz OFDM subcarriers spacing. In addition, the fourier transform scaling factor β can be set to allow different transmission bandwidths for each configuration of the subcarrier spacing. This results in the support of nominal RF bandwidth from 1,728 MHz up to 221,184 MHz.

The physical layer can perform the following functions in order to provide the data transport service:

- Error detection on the physical channel and indication to higher layers.
- FEC encoding/decoding of the physical channel.
- Hybrid ARQ soft-combining.
- Rate matching of the coded physical channel data to physical channels.
- Mapping of the coded physical channel data onto physical channels.
- Modulation and demodulation of physical channels.
- Frequency and time synchronization.
- Radio characteristics measurements and indication to higher layers.
- Multiple Input Multiple Output (MIMO) antenna processing.
- Transmit Diversity (TX diversity).
- Beamforming.

The physical channels defined are:

- the Physical Control Channel (PCC);
- the Physical Data Channel (PDC).

The modulation schemes supported are:

- BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM and 1024-QAM.

The channel coding scheme for transport blocks in all physical channels is Turbo Coding with a coding rate of R = 1/3, two 8-state constituent encoders and a turbo code internal interleaver. Trellis termination is used for the turbo coding. Before the turbo coding, transport blocks are segmented into byte aligned segments with a maximum information block size. Error detection is supported by the use of 16 or 24 bit CRC as specified for a given physical channel.

6.2.2 Radio characteristics

Radio characteristics of DECT-2020 NR combined with the physical layer supports nominal RF bandwidth from 1,728 MHz up to 221,184 MHz. The current release focuses on URLLC and mMTC use cases.

The radio channel numbering scheme enables to assign channels from 450 MHz up to 5 875 MHz organized into 17 different operating bands.

The transmitter output power can be adapted to different types of application requirements and support use cases like battery powered industrial applications, which enables the support for high equipment density use cases. The RX-TX transition time is defined to operate within the Guard Interval (GI), which enables a very competitive low latency operation with hybrid ARQ operation.

Receiver requirement defines the minimum performance for the radio device with hybrid ARQ support. The reference sensitivity levels scale depending on the operating bandwidths.

Measurement requirements are defined for channel access purposes and to support radio environment quality reporting for mobility and mesh routing purposes.

The requirements are defined keeping in mind the state of art in performance, low power consumption and competitive implementation cost.

6.3 Medium Access Control Layer

6.3.1 Introduction

The objective of this clause is to provide an overview on the MAC protocol layer architecture, the used identities, the used transport channels and the mapping of the MAC PDU into the physical layer packet.

The descriptions provided are a model of the MAC layer i.e. it does not specify or restrict implementations.

6.3.2 MAC Structure

The overall MAC structure is depicted in Figure 6.3.2-1. The MAC layer provides DTCH and MTCH logical channels for transferring higher layer data. The flow of the higher layer data and MAC internal messages to physical channels is depicted with black solid lines. The grey dashed line corresponds to MAC internal control interfaces between the MAC function and the MAC control entity.

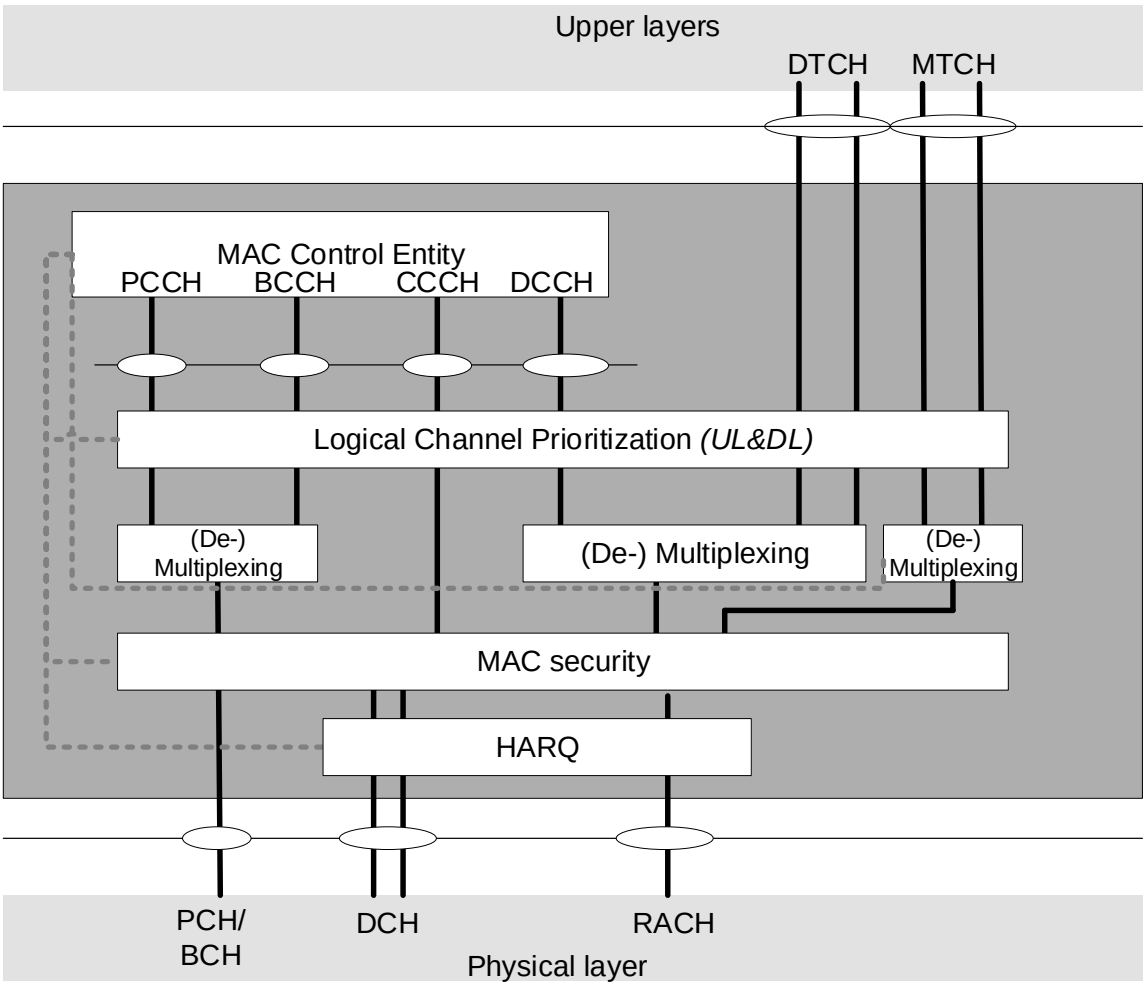


Figure 6.3.2-1: MAC structure overview

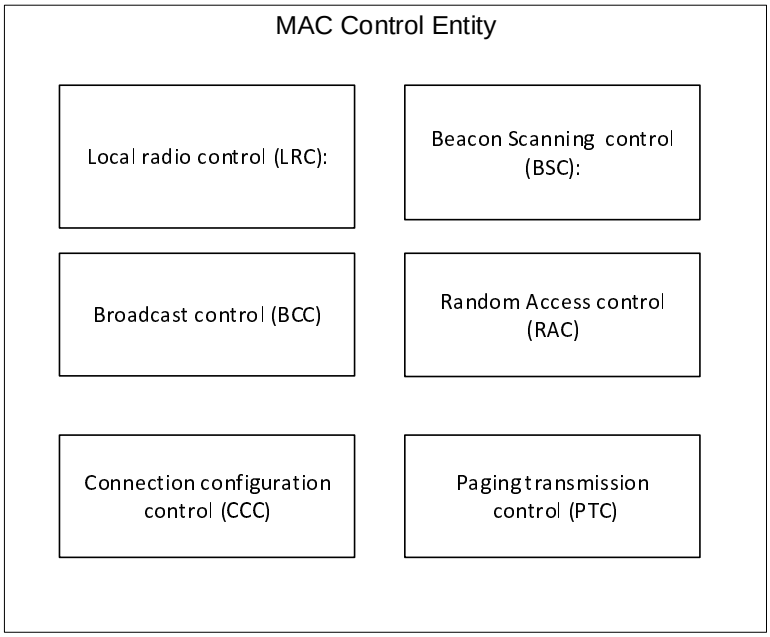


Figure 6.3.2-2: MAC control Entity

The MAC layer of the Radio Device (RD) handles transmission and reception of the following transport channels:

- Paging and Broadcast channel (PCH/BCH);
- Dedicated Channel (DCH);
- Random Access Channel (RACH).

The MAC control entity is depicted in Figure 6.3.2-2.

6.3.3 Identities

Network ID

The Network ID has a length of 32 bits, and first 24 MSB bits are used to identify a DECT-2020 network uniquely from other DECT-2020 networks. The 8 LSB bits of the network ID are selected locally to minimize collision with other networks.

The network ID is transmitted periodically in a beacon frame as plain text enabling other RDs to detect which network the transmitted beacon belongs to.

Long Radio Device ID (Long RD ID)

The long radio device ID has a length of 32 bits, and it identifies a radio device uniquely in a single DECT-2020 network. An RD obtains the long RD ID as part of the authentication process over the DECT-2020 system or via manual provision or by using another communication channel.

The long RD ID is used:

- in association procedure to recognize associating RDs uniquely;
- in MAC layer security procedures;
- in packet routing in mesh system operation to identify original source and final target receiver of the routed data packet.

The long RD ID is transmitted in the MAC PDU to:

- identify the receiver and the transmitter of the packet in the association procedure for exchanging short RD IDs; or

- when transmitter of the frame considers that there is potential confusion on short RD IDs.

Short Radio Device ID (Short RD ID)

The short radio device ID has length of 16 bits, and it identifies a radio device locally in DECT-2020 network.

The short RD ID is used in the transmitter or receiver fields of the PHY control field to identify transmitter and receiver of the packet.

The short RD ID is exchanged during association procedure between RDs performing association so that linkage between short RD ID and long RD ID is obtained by both associating RDs. Each radio device randomly selects short RD ID value that it uses in association.

6.3.4 Services

The MAC layer provides the following services to upper layers:

- data transfer;
- radio resource allocation.

The MAC layer expects the following services from the physical layer:

- data transfer services in physical layer packets;
- measurements.

6.3.5 Functions

The MAC layer supports the following control functions in the MAC control entity:

- Local Radio Control (LRC): This function is overall in charge of the radio resource in the local coordination area, when RD coordinates local radio resources, i.e. operates as FT.
- Paging Transmission Control (PTC): This function controls paging message transmission when RD coordinates local radio resources, i.e. operates as FT.
- Broadcast Control (BCC): This functionality controls Beacon and other broadcast/multicast transmissions.
- Random Access Control (RAC): This functionality is in charge of random access transmissions.
- Beacon Scanning Control (BSC): This function controls scanning operations.
- Connection Configuration Control (CCC): This functionality controls multiplexing, mapping data to transport channel, MCS, HARQ configuration, MAC security and handovers with LRC.

The MAC layer supports the following transmission functions:

- paging and broadcast signalling;
- control signalling;
- radio resource management by channel selection and channel access procedures;
- logical channel prioritization;
- mapping between logical channels and transport channels;
- multiplexing of MAC SDUs from one or different logical channels onto MAC PDU to be delivered to the physical layer on transport channels;
- demultiplexing of MAC SDUs to one or different logical channels from transport blocks MAC PDU delivered from the physical layer via transport channels;
- error correction through HARQ;

- MAC layer security by providing integrity protection and ciphering.

6.3.6 Channel Structure

The MAC layer provides data transfer services on logical channels. To accommodate different kinds of data transfer services, multiple types of logical channels are defined i.e. each supporting transfer of a particular type of information.

Each logical channel type is defined by what type of information is transferred. MAC has a set of internal logical channels and logical channels to higher layers.

The MAC layer has the control and traffic channels listed in Table 6.3.6-1.

Table 6.3.6-1: Logical channels provided by MAC

Logical channel name	Acronym	Available for higher layers	Control channel	Traffic channel
Broadcast Control Channel	BCCH		X	
Paging Control Channel	PCCH		X	
Common Control Channel	CCCH		X	
Dedicated Control Channel	DCCH		X	
Dedicated Traffic Channel	DTCH	X		X
Multicast Traffic Channel	MTCH	X		X

The MAC sublayer uses the transport channels listed in Table 6.3.6-2.

Table 6.3.6-2: Transport channels used by MAC

Transport channel name	Acronym	Downlink	Uplink
Paging and Broadcast Channel	PCH/BCH	X	
Dedicated Channel	DCH	X	X
Random Access Channel	RACH	X	X

Logical channels can be mapped as described in Table 6.3.6-3.

Table 6.3.6-3: Channel mapping

Transport channel / Logical channel	PCH/BCH	DCH	RACH
BCCH	X		
PCCH	X		
CCCH			X
DCCH	X	X	X
DTCH		X	X
MTCH	X		X

In packet transmission transport channels are mapped to Physical Data Channel (PDC) carried in the physical layer packet. Additionally, MAC provides the number of used spatial streams, and content of the Physical Layer Control Field to the physical layer. The Physical Layer Control Field is mapped to the Physical Control Channel (PCC) of the physical layer packet.

6.4 Mobility and State Transitions

6.4.1 Overview

The RD_{PT} operates in the following states: non-associated and associated. The mobility support, i.e. handover, is applicable for an RD which is associated to one or more RD_{FT} or $RD_{FT,PT}$. The different handover procedures are introduced in clause 6.4.2, wherein the handover functionality is the same in both star and mesh topologies.

6.4.2 Intra DECT-2020 Mobility

The involved handover procedures may be intra-cell or inter cell. The intra-cell handover means RD_{PT} or $RD_{FT,PT}$ changes its association from an instance (e.g. sector) of an RD_{FT} or $RD_{FT,PT}$ into another instance of the same RD_{FT} or $RD_{FT,PT}$. The inter-cell handovers consider changing association from an RD_{FT} or $RD_{FT,PT}$ into another RD_{FT} or $RD_{FT,PT}$.

Intra-cell handover may be controlled by either the radio devices acting as RD_{FT} or RD_{PT} and triggered when the quality on allocated resources becomes poor and other free resources exist. Detection of free resources is based on a spectrum sensing and beacon measurements and takes into account the activity of other uncoordinated systems. Seamless handover is supported, by allocating new resource before releasing old ones.

The radio device sends a handover-request (part of association request process) to the detected RD_{FT} on the selected random-access channel indicated in beacon message. If the RD_{FT} accepts the request, it indicates the position of the new resources and the connection is established, and data delivery can be switched over. After that the old resources will be released.

Inter-cell handover is generally controlled by the RD_{PT} and triggered when quality on allocated resources becomes poor and/or another suitable RD_{FT} is detected. Seamless handover with 0 ms interruption time is supported. The RD_{PT} sends a handover-request as indicated in intra-cell case to the new RD_{FT} on the selected random access channel. If the RD_{FT} accepts the request, then it indicates the position of the new resource allocation and the data will be switched over. After that the old resource will be released.

Handover between two DECT-2020 systems is performed in the same way as inter-cell handover. Both systems should be interconnected by the proper network infrastructure.

History

Document history		
V1.1.1	July 2020	Publication
V1.2.1	April 2021	Publication